AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 18 · STUDENT LESSON

Confidence Is Not Correctness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Read the claim. Check the outcomes. Keep the unknowns.

Interpret score meanings, compare probability claims with evidence and audit incomplete result reports.

Keep one habit: Define what the score claims → check the named event → retain assessed and missing counts.

MAIN LESSON · SECTION 1

What does the confidence claim actually mean?

> Read the score definition, check outcomes and retain missing evidence before judging the claim.

Start with tone and correctness

A certain-sounding answer can be wrong, and a cautious answer can be right. Keep the confidence claim's meaning separate from the evidence used to check the prediction.

Retrieve the missing-outcome question

A **prediction** says far and sounds very certain. No travel distance has been recorded. Can you mark it correct? Explain what is missing.

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can distinguish a **confidence claim** from **evidence** that a **prediction** was correct.
- I can compare recorded **confidence claim**s with observed outcomes for a group.
- I can explain why a score, a confident sentence and repeated agreement need different checks.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

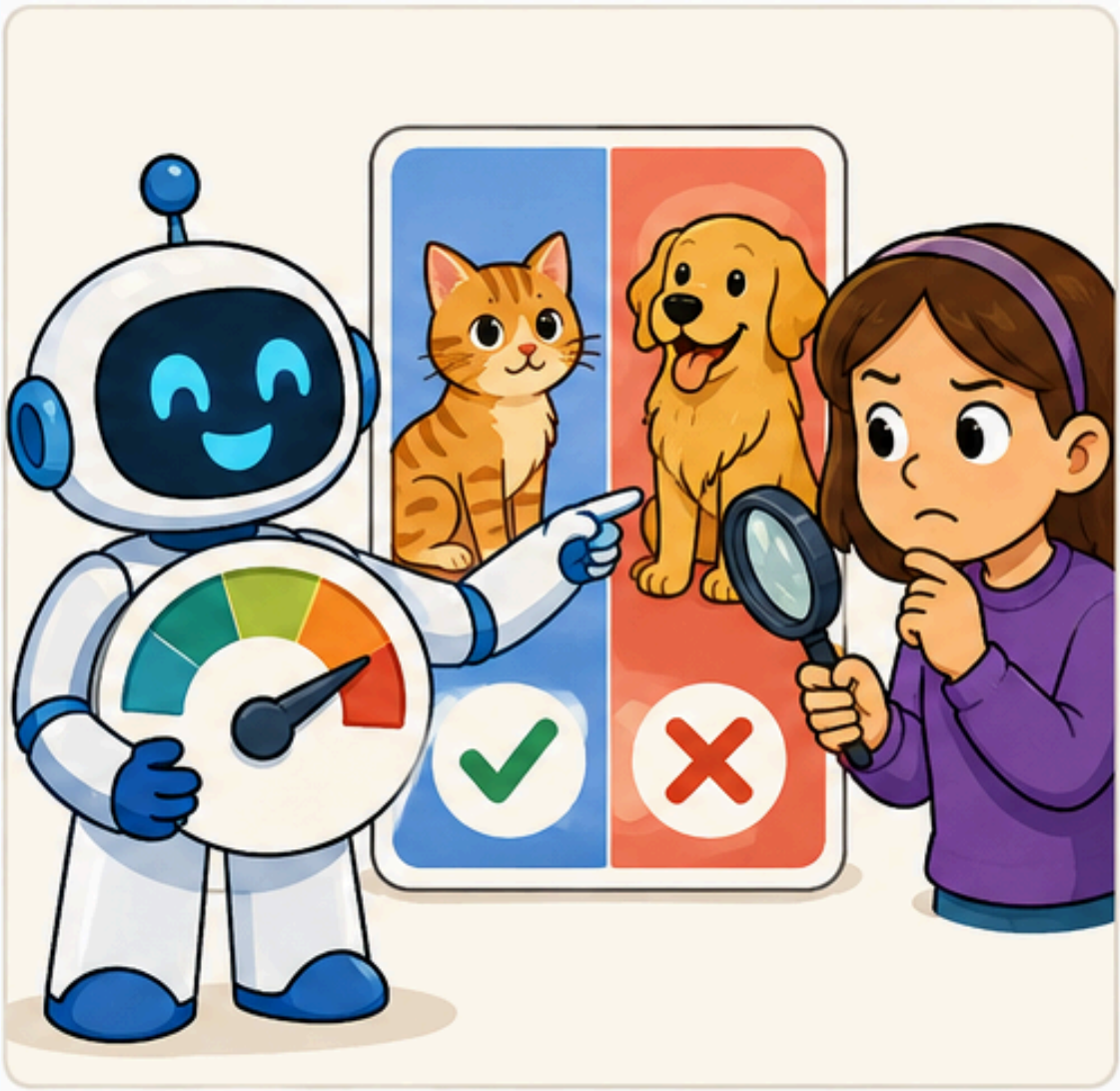
Remember before reading

Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

A system can be very confident and still be wrong.



LEARNING TARGET A helper says 'definitely correct' on four answers; only two match the answer key. A cautious helper gives the same four answers. Which is more accurate?

What additional **evidence** would you need to compare **confidence** quality?

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can explain why a **confidence** score is not a guarantee and account for missing outcomes.

MAIN LESSON · SECTION 2

Read the definition before interpreting the number

> A score, a probability claim and confident wording can describe different things.

Separate the claim and its outcome check

A **prediction** can be expressed strongly and still be wrong. **Confidence** describes a claim about support or likelihood; **correctness** is checked against **evidence**.

Keep the **prediction**, the **confidence claim** and the observed outcome in separate columns. None can replace the others.

Inspect the meaning of an 80 display

A display reading 80 might mean 80 out of 100 points for matching features, an estimated 80 percent chance of a specified event, or something else. Read its definition before interpreting it.

The number alone does not tell you which meaning applies. A sentence such as definitely far is not a measured probability.

Keep invented probabilities out of Ramp1

Ramp1 from Week 17 produced categories using a threshold. It supplied no **probability estimate**s. We must not attach a probability merely because the height is far from that threshold.

This week uses fictional forecast records with an explicitly stated probability claim; we are checking those claims, not inventing them for Ramp1.

Explain what each claim requires

- What event does a probability estimate refer to? How would you distinguish an 80-point matching score from an estimated 80 percent chance of correctness? Does Ramp1 supply either probability estimate?

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A classifier assigns 95 percent **confidence** to 'dog' for a toy fox but the verified label is fox.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

MAIN LESSON · SECTION 3

Compare the probability claim with groups of outcomes

> A finite group supplies evidence without guaranteeing an exact future count.

Read three different evidence situations

Claims and results are separate records

BEFORE OUTCOMES	AFTER OUTCOMES	WHAT THE GROUP SHOWS
Ten claims of 80 percent	Eight predictions correct	8 out of 10 matched
Ten claims of 80 percent	Five predictions correct	5 out of 10 matched
One claim of 80 percent	Outcome not available	Correctness unknown

Here each percentage claims the probability that its chosen category will be correct. The two ten-case groups are fictional examples. A group comparison cannot identify an unknown individual outcome.

Explain what an 80 percent claim estimates

An 80 percent claim means an estimated chance of 80 out of 100, for the event that was named. Across many relevant cases carrying that claim, a well-calibrated method should be correct about that often.

It does not promise exactly eight successes in every ten cases or tell us which cases will fail.

Compare both observed groups within their limits

Eight correct out of ten agrees numerically with 80 percent in this small group. Five out of ten is lower. Both are **observation**s worth reporting; neither small group alone establishes how dependable the method will be.

Larger relevant checks, independent of model **fitting** and score adjustment, give a stronger basis for judging the claim.

Explain one comparison

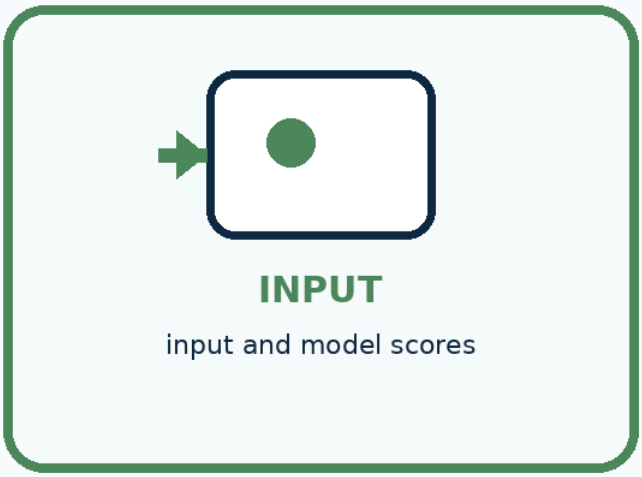
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

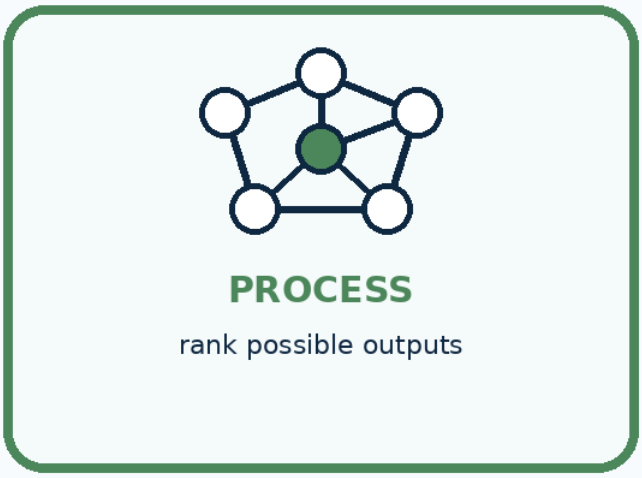
▼ **Earlier wording, alternatives and picture transcripts**

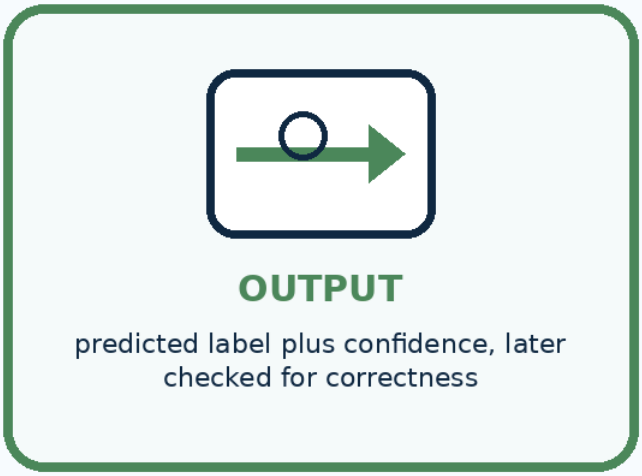
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT input and model scores

PROCESS rank possible **output**s

OUTPUT predicted label plus **confidence**, later checked for **correctness**

CORE IDEA A system can be very confident and still be wrong.

UNDER THE HOOD **Confidence** is model-relative strength, not a calibrated probability of truth unless validated. Distribution shift, bad labels, missing classes, or over**fitting** can produce high-**confidence** errors.

Draw a fresh example with all three stages:

A closer explanation

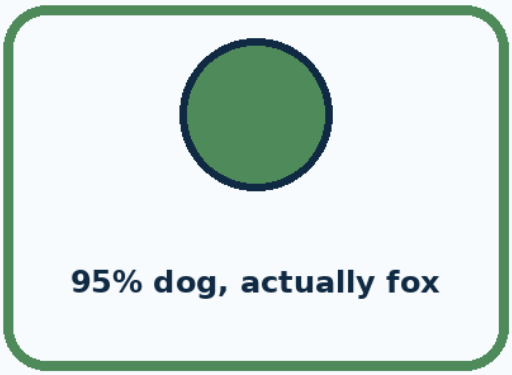
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare confident errors and cautious successes

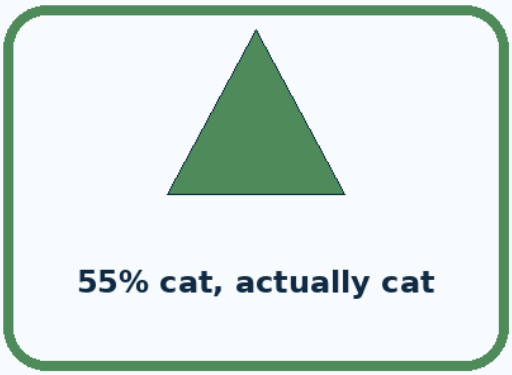
> Use checked outcomes to judge correctness and definitions to interpret confidence claims.

Pictures paired with their evidence



The case to investigate

CASE 1 — 95% dog, actually fox Clue: strong wrong **output**. Meaning: **confident error**.



The case to investigate

CASE 2 — 55% cat, actually cat Clue: weak correct **output**. Meaning: cautious correct.



The case to investigate

CASE 3 — 50/50 tie Clue: competing **output**s. Meaning: uncertain.



The case to investigate

CASE 4 — Unknown animal Clue: class absent. Meaning: **confidence** may mislead.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

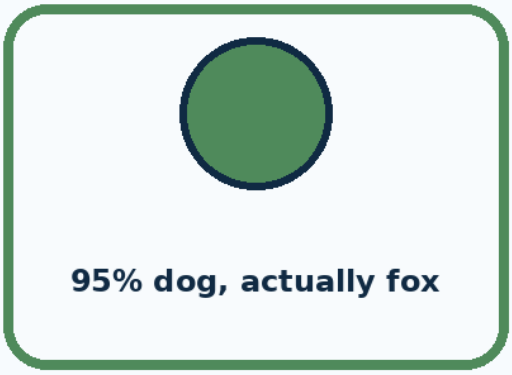
BOUNDARY Avoid presenting a score as an objective certainty when its meaning has not been validated.

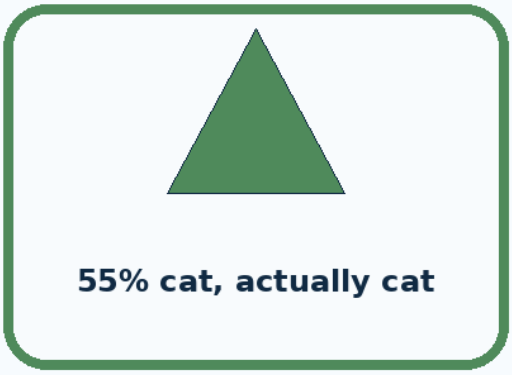
▼ **Earlier wording, alternatives and picture transcripts**

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — 95% dog, actually fox Clue: strong wrong **output**. Meaning: **confident error**.

CASE 2 — 55% cat, actually cat Clue: weak correct **output**. Meaning: cautious correct.

CASE 3 — 50/50 tie Clue: competing **output**s. Meaning: uncertain.

CASE 4 — Unknown animal Clue: class absent. Meaning: **confidence** may mislead.

Change one thing in the pictured case

Change: The missing outcome is checked and found incorrect.

Prediction: All-ten agreement is 7/10=**70%**.

Reason: There are now three wrong labels and no unknown outcomes.

Keep the missing outcome in the report

> The ten selected cards have seven established correct outcomes, two incorrect and one unavailable.

The defined claim and selected-case evidence

A fictional card classifier reports a claimed **80%** chance that each chosen label is correct.

Ten **prediction**s all have score **80%**.

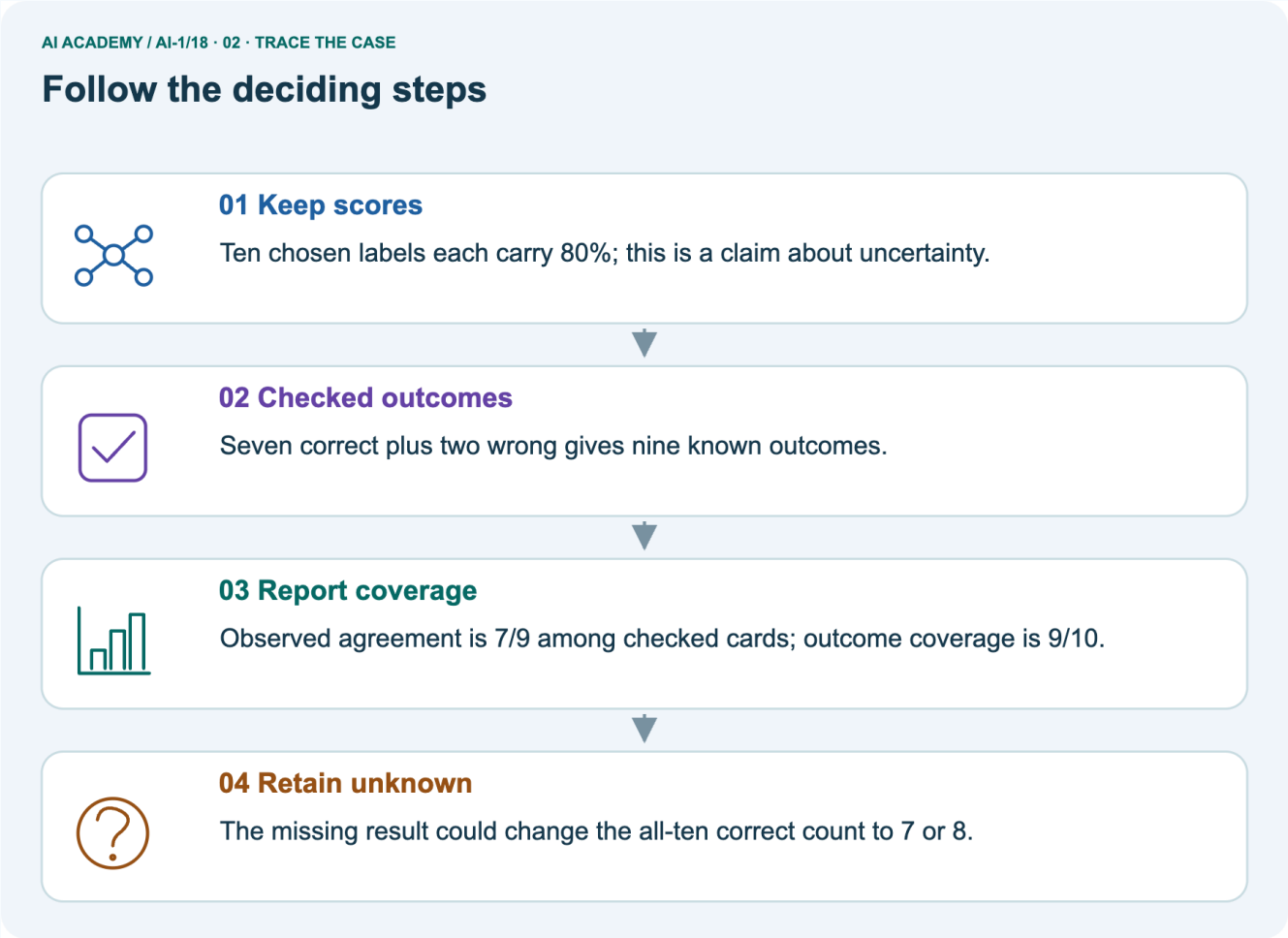
Later checks find 7 correct, 2 incorrect and 1 outcome unavailable. The selected cards were not a representative survey.

Read known outcomes and missing coverage



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the assessed and all-ten counts



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the possible overall range

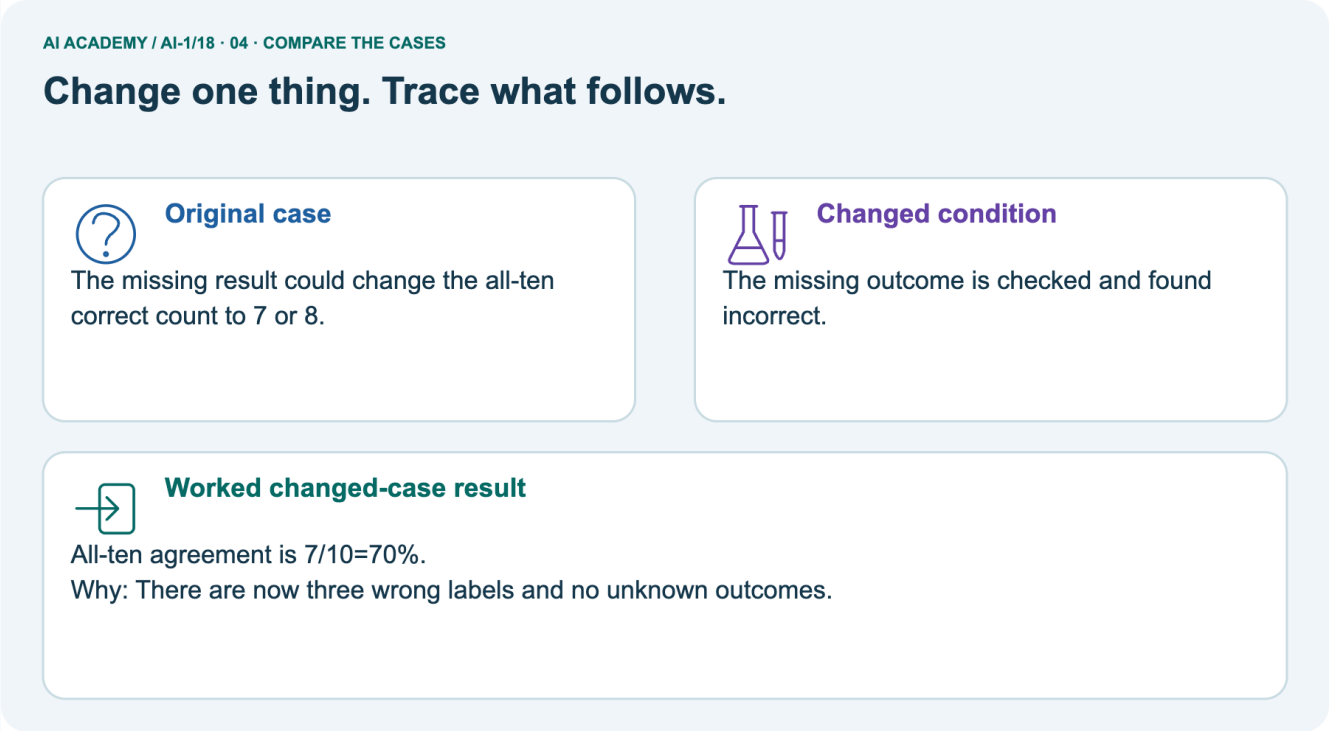
Do not count the missing outcome as correct or silently remove it from the coverage report. These records allow an all-ten agreement range from **70%** to **80%**.

They do not show that claimed probabilities match checked **correctness** over many comparable **prediction**s; that requires much more suitable **evidence**.

Predict after the missing result is checked

- Before checking, suppose the one missing outcome is now established as incorrect. Keep every original 80 percent claim unchanged. Recount correct, incorrect, assessed and unknown cases, then state the all-ten agreement fraction.

▼ Check the completed count



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit claims about calibration and future cards

This small selected set cannot establish that the claimed probabilities match actual **correctness** or show performance for all future cards.

Explain why the missing label was never guaranteed

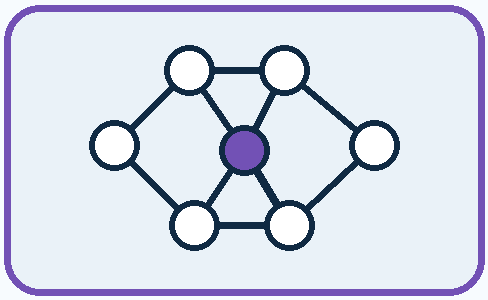
- Read the labelled picture: Was the unknown label guaranteed correct because its **confidence** was **80%**?

▼ **Earlier wording, alternatives and picture transcripts**

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A classifier assigns 95 percent **confidence** to 'dog' for a toy fox but the verified label is fox.

STEP 1 — OBSERVE **input** and model scores

STEP 2 — TRACE rank possible **output**s

STEP 3 — CONCLUDE predicted label plus **confidence**, later checked for **correctness**

MODEL REASONING **Confidence** and **correctness** are separate axes. Only verified outcomes establish **correctness**. Compare many cases to learn whether **confidence** is trustworthy; one 95 percent value is not proof.

DO NOT OVERCLAIM Avoid presenting a score as an objective certainty when its meaning has not been validated.

What new **evidence** could change this answer?

A closer explanation

Case and recorded claim	Observed outcome	Judgment
A predicts far at 90 percent	Verified near	Incorrect despite high confidence
B predicts near at 60 percent	Verified near	Correct despite lower confidence
C predicts far at 90 percent	Outcome missing	Unassessed

A high-**confidence** error and a lower-**confidence** success can occur even when a probability method is useful. One error does not prove that all **confidence** estimates are worthless, and one success does not validate a method.

Keep the original scores and investigate patterns across relevant cases.

Record the task, model version, event being assigned a probability, score definition and **evidence** used for checking. If outcomes are missing, report that count.

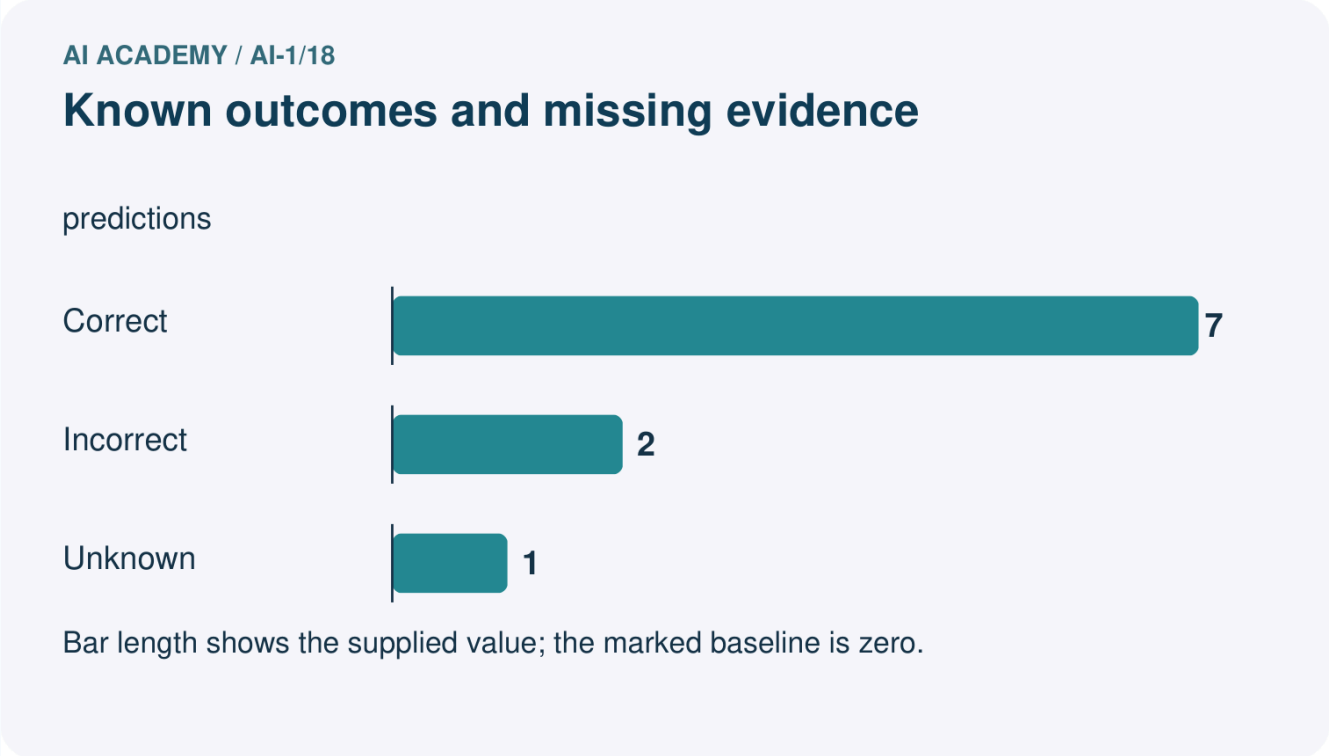
Do not quietly remove difficult cases or change the **confidence** values after reading the answers. Adjustments belong to a new recorded method and need a fresh evaluation.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Confidence Is Not Correctness

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/18

Confidence Is Not Correctness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep scores

Ten chosen labels each carry 80%; this is a claim about uncertainty.



2. Checked outcomes

Seven correct plus two wrong gives nine known outcomes.



3. Report coverage

Observed agreement is 7/9 among checked cards; outcome coverage is 9/10.



4. Retain unknown

The missing result could change the all-ten correct count to 7 or 8.

CHANGE ONE THING

The missing outcome is checked and found incorrect.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep scores	Ten chosen labels each carry 80%; this is a claim about uncertainty.
2. Checked outcomes	Seven correct plus two wrong gives nine known outcomes.
3. Report coverage	Observed agreement is 7/9 among checked cards; outcome coverage is 9/10.
4. Retain unknown	The missing result could change the all-ten correct count to 7 or 8.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: The missing outcome is checked and found incorrect. Predict the result before checking the explanation. What remains the same?

Seek evidence instead of stronger wording

> Read the score definition and request relevant outcome checks.

Ask what the display measures and how it was checked

Before relying on a **confidence** display, ask what its number means, how it was checked on relevant cases and what happens when it is wrong. A feature-match score may help rank options without being a probability.

A generated sentence may sound certain without supplying either a defined score or supporting **evidence**.

Keep repeated agreement and permission separate from truth

Repeating a claim does not establish its truth. Several tools can share information and mistakes. Useful checks seek relevant independent **evidence** and preserve unresolved outcomes.

Review decisions also depend on the consequences of an error; a probability alone does not grant permission to act.

Repair the invented 100-percent claim

- CLAIM A Higher **confidence** always means more correct.
- CLAIM B A correct low-**confidence** answer proves **confidence** is useless.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **A tool says definitely correct and repeats the same answer five times. A learner writes 100 percent confidence and skips the outcome check. Explain two unsupported steps and name evidence that would be more useful**

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
95% dog, actually fox	strong wrong output	confident error
55% cat, actually cat	weak correct output	cautious correct
50/50 tie	competing outputs	uncertain
Unknown animal	class absent	confidence may mislead

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: report a checked group without guaranteeing the next case

> Keep the probability event, original claim and observed counts visible.

The separate fully checked ten-case group

A fictional classifier records an 80 percent probability that each selected category will be correct on ten reserved cases. All ten outcomes are later established: eight correct and two incorrect.

The report should preserve both the claim and the observed counts. The group supports a comparison, not a guarantee about the next case.

> **Write a short report for this ten-case group. State the event claimed, the recorded probability, both outcome counts and one limitation. Explain why the next prediction at 80 percent cannot be marked correct before its outcome is checked**

Retain missing outcomes when checks are incomplete

Now suppose two of the ten outcomes were missing. You would need to report which cases were assessed and how many remained unknown, as in Week 17.

A confident-looking percentage does not repair missing **observation**s or make an incomplete sample representative.

Use W2 to judge cases from their outcomes

Use workbook W2 for the connected example "**Confidence** and **correctness**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “80 points always means 80 percent probability”

- Use the worked case for **Confidence** and **correctness**. Proposed explanation: “An 80 point score always means an 80 percent probability”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the pictured unknown outcome

Was the unknown label guaranteed correct because its **confidence** was **80%**?

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A system can be very confident and still be wrong.

Use a fresh example to explain **confidence**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: A correct low-**confidence** answer proves **confidence** is useless.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Judge cases using outcomes

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate a confidence-and-correctness grid

> Use complete records that distinguish strong wording, caution, a probability claim and a matching score.

Your four supplied records

Use four fictional helper records. A says “Definitely, 2+3=6”; B says “Maybe, 2+3=5”. Check both against arithmetic. C predicts a card is red with a stated estimated 75 percent chance of that prediction being correct; the supplied check records red. D predicts circle with match score 80 out of 100 feature points, explicitly not a probability; no outcome check is available.

Check outcomes and explain each claim type

1. For each record, write the prediction, confidence wording or score definition, evidence used for checking, and correct/incorrect/unassessed status. Explain why A’s tone cannot rescue its arithmetic, B’s cautious wording cannot make it incorrect, C’s one success cannot validate its probability method, and D still needs an outcome. Do not convert “maybe” into an invented percentage.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep the evidence and limits

Case	Prediction	Observed result	What it means
Normal			
Edge			

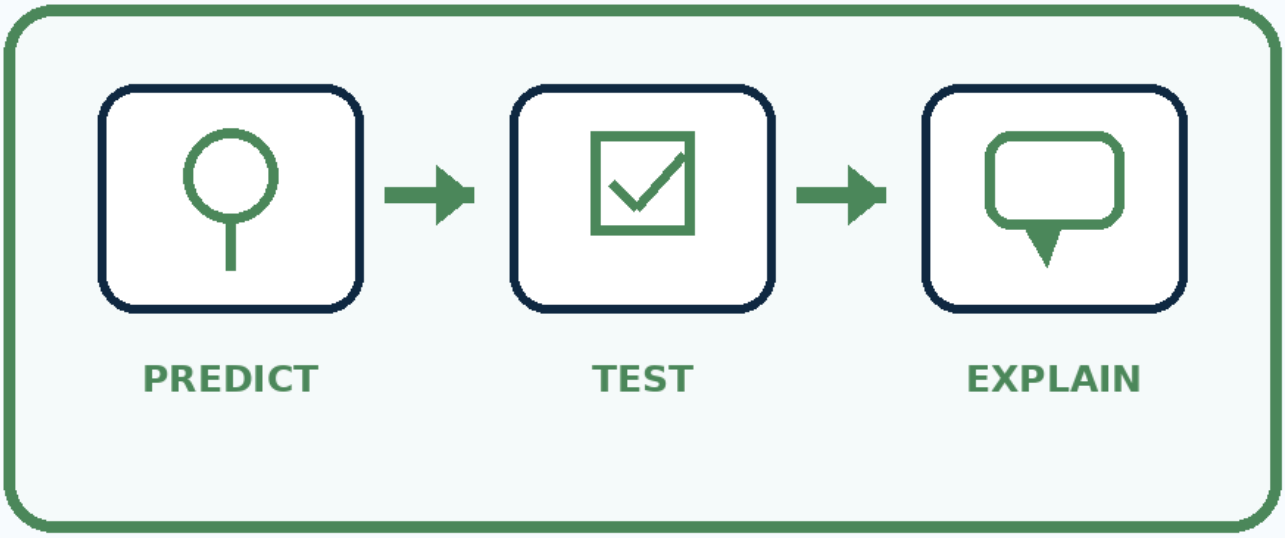
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EXPLORER LAB

Confidence-Correctness Grid

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Compare **confident error**s with cautious correct answers.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit the leaf-sorter probability claim

> Keep the eight chosen cases, seven established outcomes and one unknown distinguishable.

The new claim and report to repair

A fictional leaf sorter claims a 75 percent probability of **correctness** for each of eight chosen labels. Later **evidence** establishes six correct labels, one incorrect label and one missing outcome.

A report says six out of eight proves perfect **calibration** and guarantees the next label. Repair the report, state the assessed and unassessed counts, and describe a better follow-up check.

> **My independent explanation, trace and limits**

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Audit a different probability claim

Week 18 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A sorter is very confident that a leaf is a coin, but a person checks and sees a leaf. Which determines **correctness**: **confidence** or the checked object? Why?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Record both the sorter's **confidence** wording and the human-confirmed outcome; keep them in separate columns.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A helper says 'definitely correct' on four answers; only two match the answer key. A cautious helper gives the same four answers. Which is more accurate? What additional **evidence** would you need to compare **confidence** quality?

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep sorter scores beside separate reference outcomes

> Record missingness and use a review policy instead of treating a high score as a fact.

This week’s project decision

THIS WEEK'S CHECKPOINT Record both the sorter's **confidence** wording and the human-confirmed outcome; keep them in separate columns.

Keep sorter scores, reference outcomes and missingness separately; use review policies instead of treating high scores as facts.

Plan score definitions, checks and review

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Keep claims tied to validated meaning

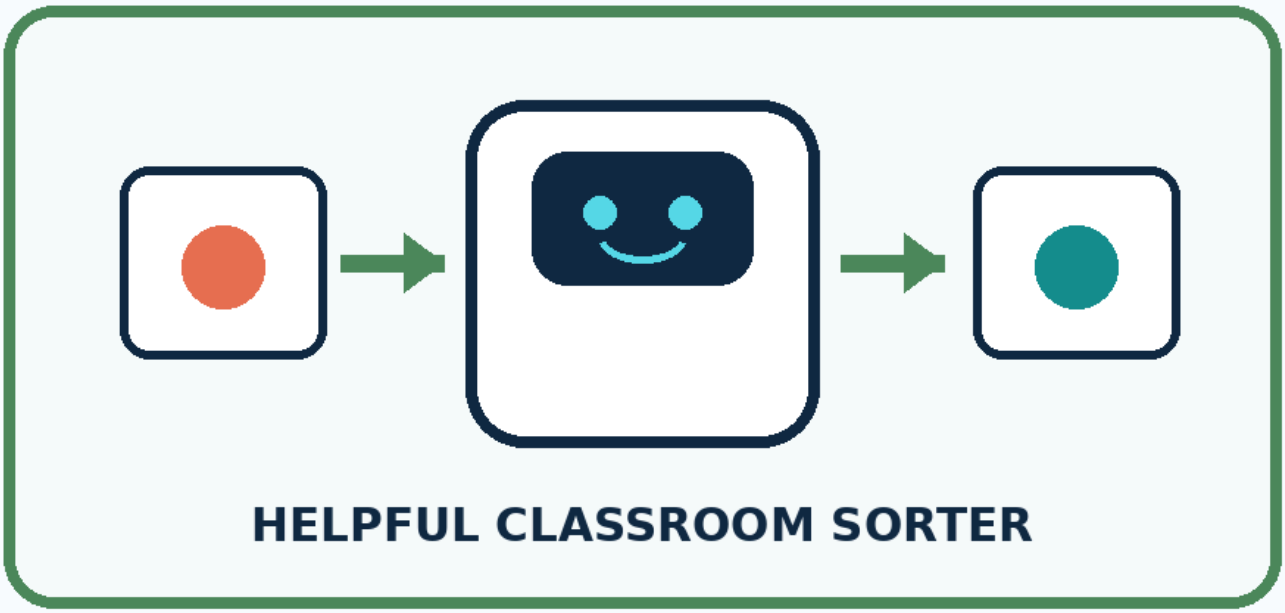
RESPONSIBLE DESIGN Avoid presenting a score as an objective certainty when its meaning has not been validated.

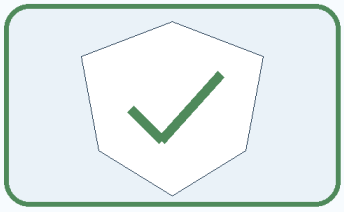
▼ Earlier wording, alternatives and picture transcripts

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: state the claim and its checking evidence

> Close your notes and explain a missing outcome, an assessed fraction and one honest limit.

Your exit explanation

- Close your notes. Explain this goal in your own words: Distinguish a **confidence claim** from **evidence** that a **prediction** was correct

My deciding step, evidence and one remaining question

Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
confidence	strength assigned to an output by a system	
correctness	agreement with verified outcome	
calibration	how well confidence matches observed accuracy	
confident error	wrong result with strong stated confidence	

Terms for the connected explanation

Word	Meaning
Confidence claim	A statement or number expressing how strongly a prediction is supported according to a stated method
Probability estimate	A numerical estimate of how likely a specified event is

Word	Meaning
Score meaning	The definition explaining what a reported number measures

Repair a claim using a word

- CLAIM TO REPAIR Higher **confidence** always means more correct.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

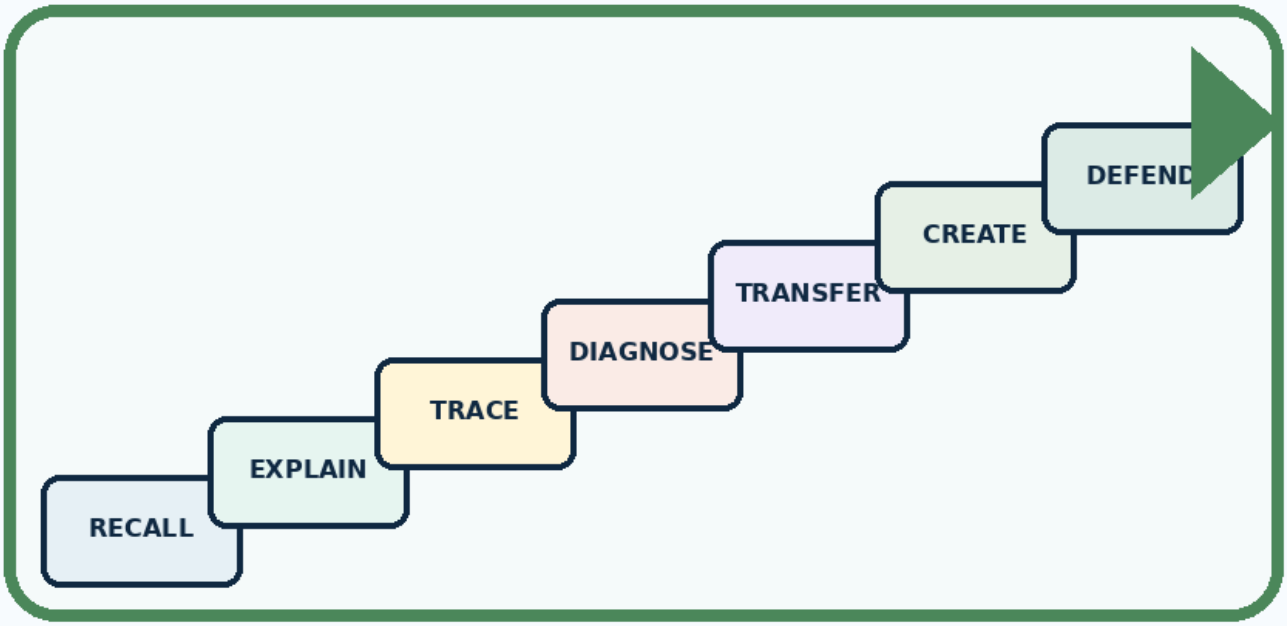
- RECALL Define **confidence**.
- EXPLAIN A system can be very confident and still be wrong.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: Higher **confidence** always means more correct.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Audit a different probability claim is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Audit a different probability claim is the selected independent case.



EXPLANATION CHECK Read the score's definition, keep original claims and checked outcomes separately, report assessed and unknown counts and avoid a guarantee from a small selected group.

Independent **evidence**:

SELF-CHECK **Confidence** and **correctness** are separate axes. Only verified outcomes establish **correctness**. Compare many cases to learn whether **confidence** is trustworthy; one 95 percent value is not proof.

NEXT BRIDGE Carry today's **evidence** habit into: Similarity.

▼ **Extra practice: Confidence cannot replace evidence**

REFERENCE · SECTION 13

Extra practice: Confidence cannot replace evidence

> Try the question before opening its worked explanation.

Read the supplied case and question

A helper confidently labels a blurred dog photo as cat; the clear original shows a dog.

What does this show?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Confidence and **correctness** differ. A **confident error** remains wrong. Check **evidence**, particularly unfamiliar or low-quality **input**s.

Evidence or stage	Value or result
Blurred image	Helper confidently says cat
Clear original	Actually a dog
Conclusion	Confident prediction was wrong

▼ Earlier wording, alternatives and picture transcripts

Week 18 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Confidence cannot replace evidence

A helper confidently labels a blurred dog photo as cat; the clear original shows a dog.

What does this show?

Confidence and **correctness** differ. A **confident error** remains wrong. Check **evidence**, particularly unfamiliar or low-quality **input**s.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Distinguish a confidence claim from evidence that a prediction was correct	Explain the deciding step, show a trace or test, and state one limit.
Compare recorded confidence claims with observed outcomes for a group	Explain the deciding step, show a trace or test, and state one limit.
Explain why a score, a confident sentence and repeated agreement need different checks	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional card classifier reports a claimed **80%** chance that each chosen label is correct.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional card classifier reports a claimed **80%** chance that each chosen label is correct.
- Which labelled step matches this detail? The missing result could change the all-ten correct count to 7 or 8.
- 1. Checked outcomes
- 2. Report coverage
- 3. Retain unknown
- 4. Keep scores
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Retain unknown The missing result could change the all-ten correct count to 7 or 8.

Do not count the missing outcome as correct or silently remove it from the coverage report. These records allow an all-ten agreement range from **70%** to **80%**.

They do not show that claimed probabilities match checked **correctness** over many comparable **prediction**s; that requires much more suitable **evidence**.

Changed-case question

- Changed-case practice: The missing outcome is checked and found incorrect. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. All-ten agreement is 7/10=**70%**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: All-ten agreement is 7/10=**70%**. There are now three wrong labels and no unknown outcomes.

Repair and save your work

Review **Prediction** Is Not Fact (ai-1/17). Goal: Separate a recorded **prediction** from the **observation** used to check it. Short practice: Which field changes the **observation** label, and which field changes the model **prediction**?.

Return to this week and explain what you changed.

Review Save New Examples for Testing (ai-1/15). Goal: Separate examples used to choose a model from examples reserved for its final evaluation. Short practice: Does 4/4 on the replay replace the original 3/4 result? Explain..

Return to this week and explain what you changed.

Review Debug the Cause (ai-1/12). Goal: Compare expected and observed traces to locate the first difference. Short practice: Propose **input** 0 and give both predicted **output**s without inventing an observed result..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

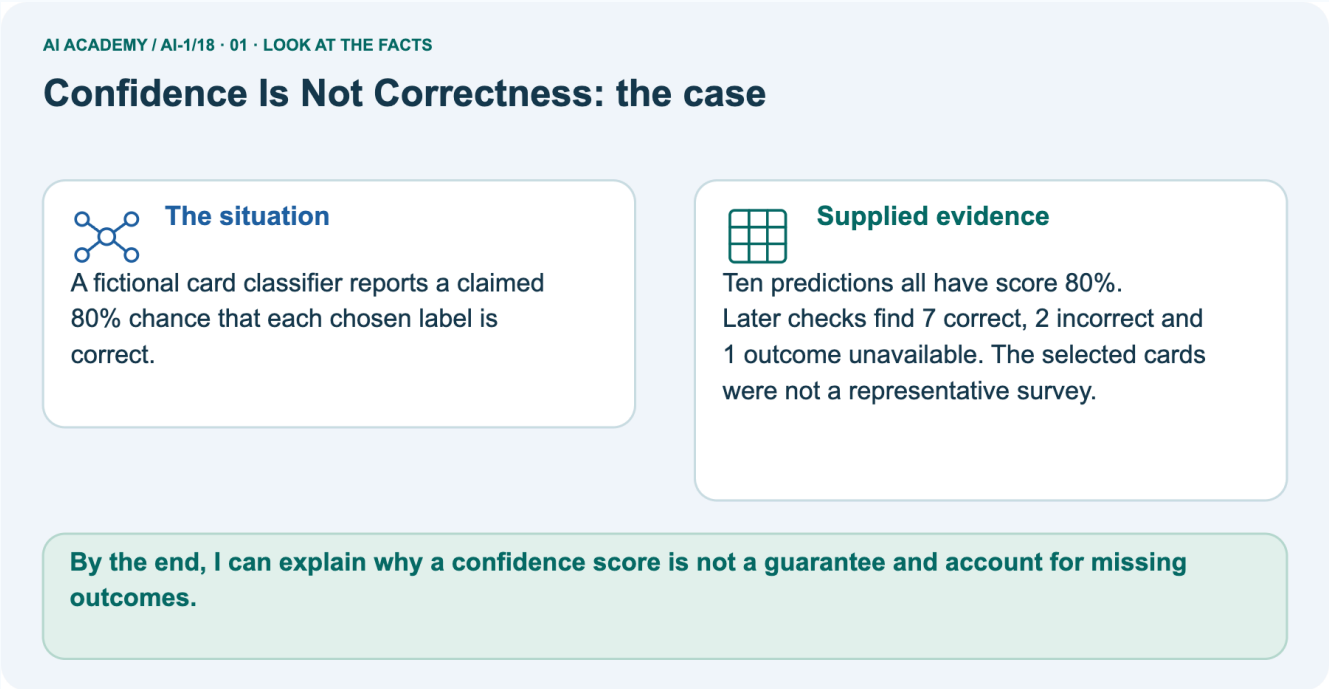
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Confidence Is Not Correctness: the case



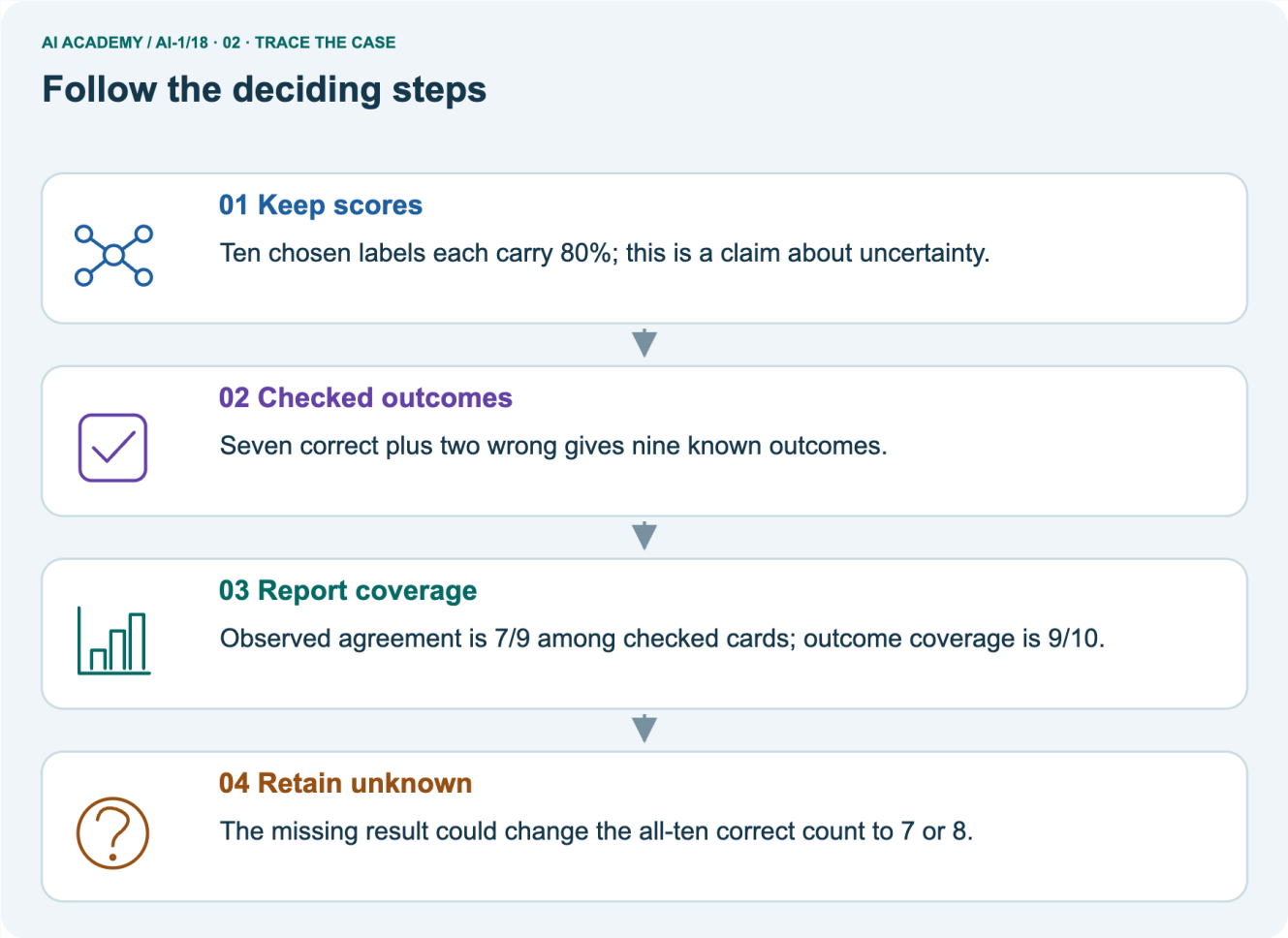
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional card classifier reports a claimed **80%** chance that each chosen label is correct. Supplied facts: Ten **prediction**s all have score **80%**. Later checks find 7 correct, 2 incorrect and 1 outcome unavailable.

The selected cards were not a representative survey.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep scores: Ten chosen labels each carry **80%**; this is a claim about uncertainty. → Checked outcomes: Seven correct plus two wrong gives nine known outcomes. → Report coverage: Observed agreement is 7/9 among checked cards; outcome coverage is 9/10. → Retain unknown: The missing result could change the all-ten correct count to 7 or 8.

Point to each numbered stage and explain which recorded **input** it uses.

03. Known outcomes and missing evidence

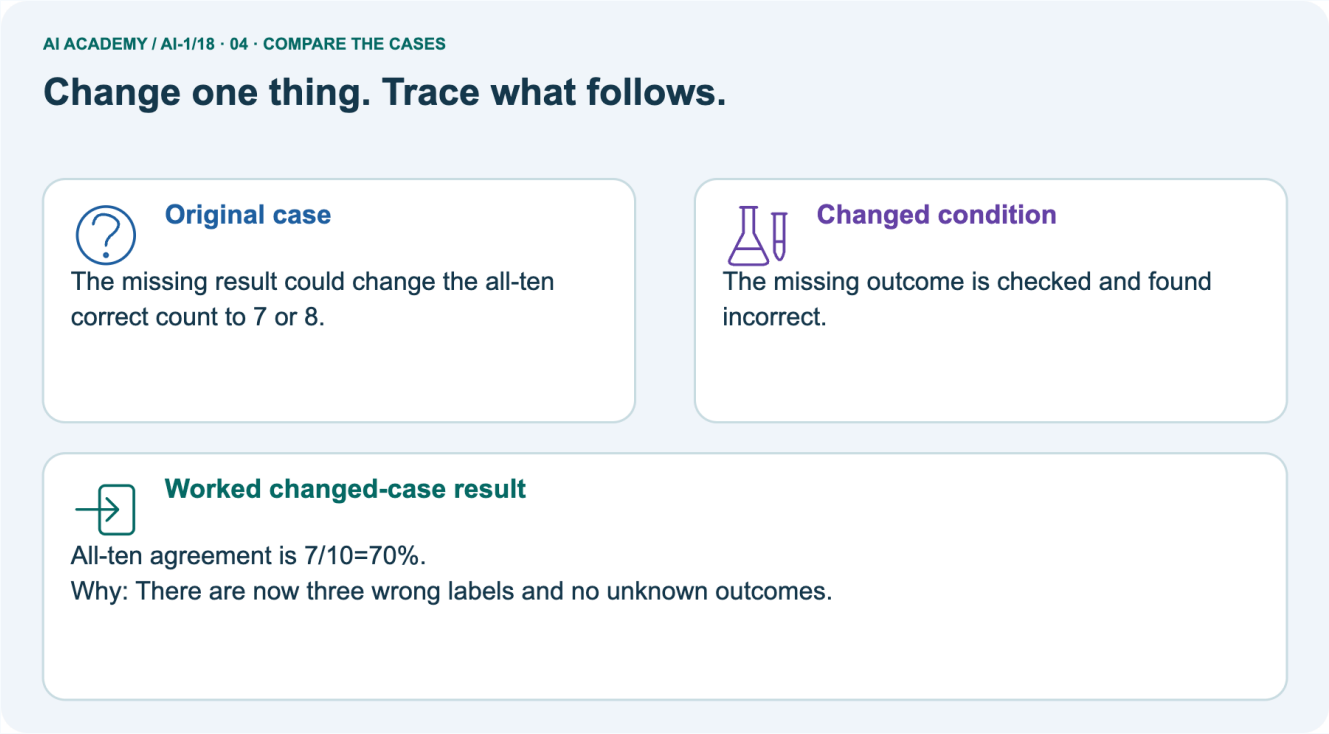


Follow the labelled picture with the matching explanation. Open the vector diagram

Known outcomes and missing **evidence**. Units: **prediction**s. Correct: 7. Incorrect: 2. Unknown: 1.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



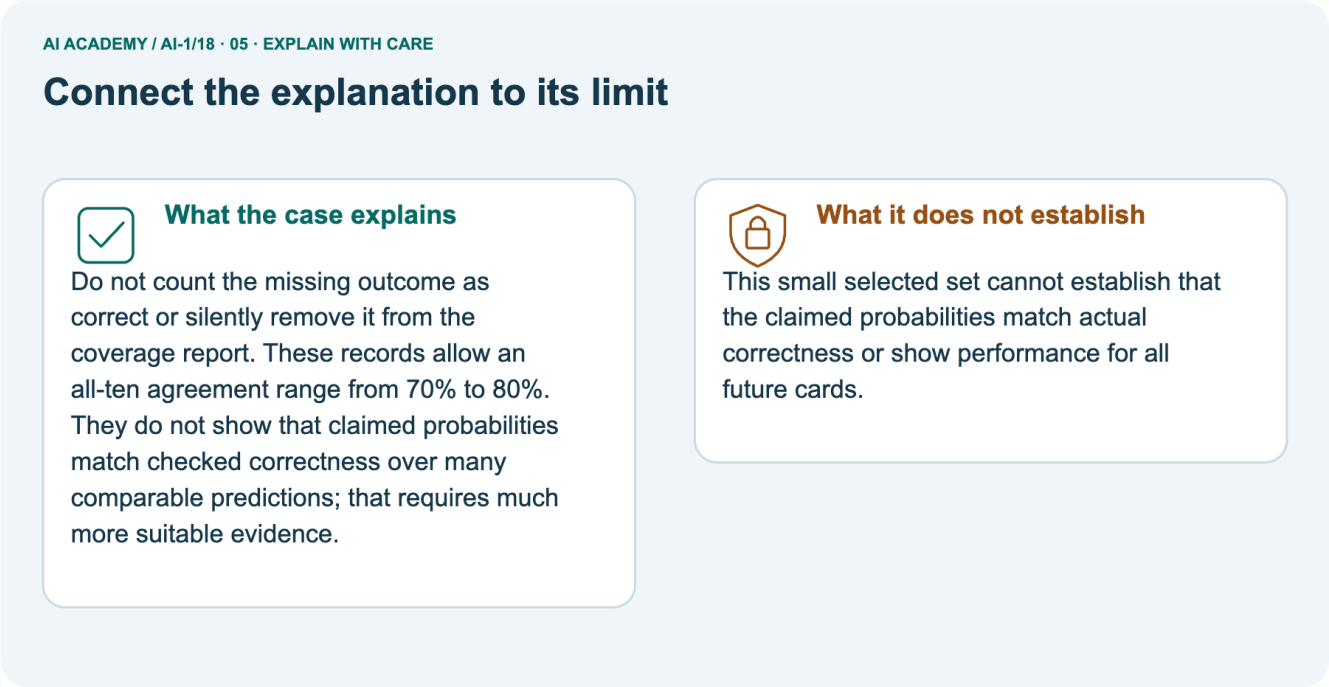
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The missing result could change the all-ten correct count to 7 or 8. Changed condition: The missing outcome is checked and found incorrect. Expected result: All-ten agreement is 7/10=**70%**.

Reason: There are now three wrong labels and no unknown outcomes.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

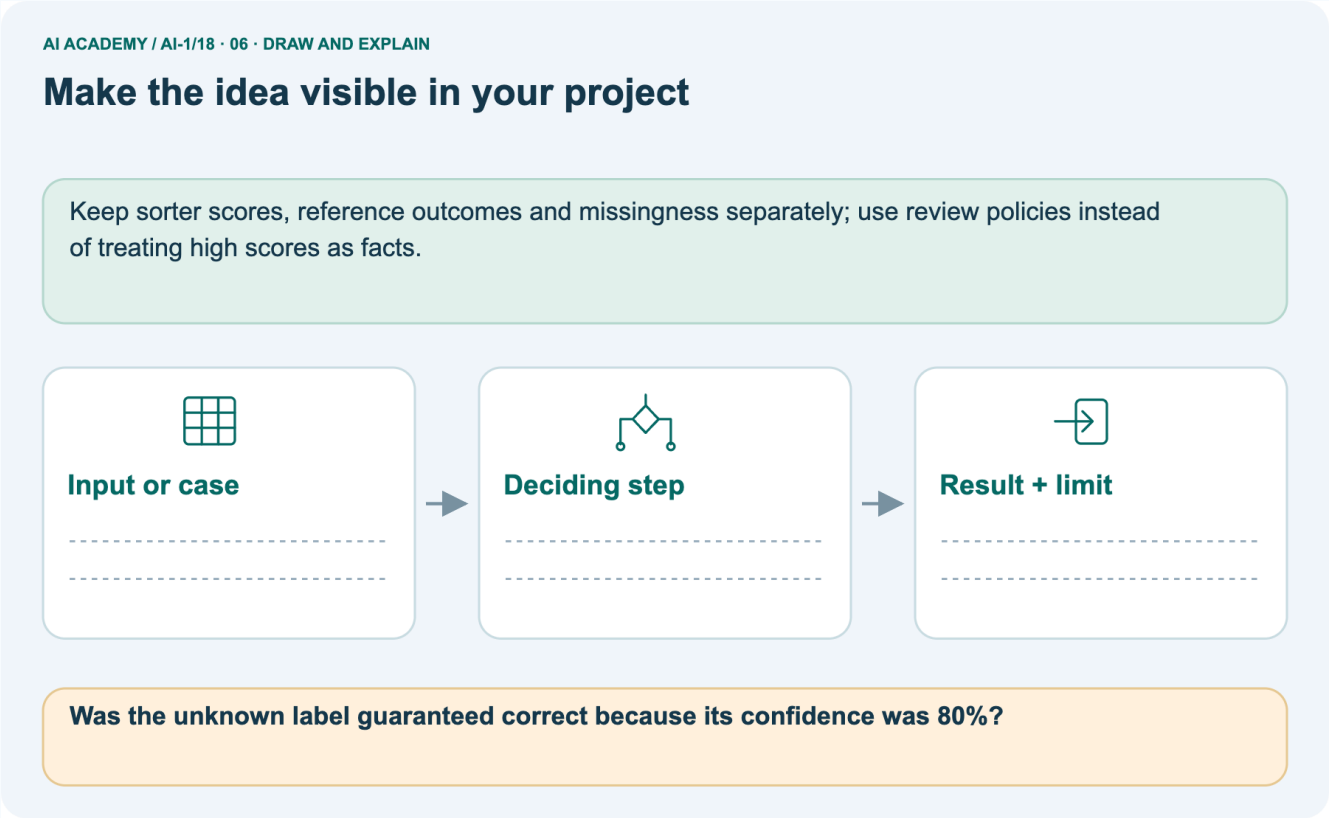
Do not count the missing outcome as correct or silently remove it from the coverage report. These records allow an all-ten agreement range from **70%** to **80%**.

They do not show that claimed probabilities match checked **correctness** over many comparable **prediction**s; that requires much more suitable **evidence**.

Limit: This small selected set cannot establish that the claimed probabilities match actual **correctness** or show performance for all future cards.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep sorter scores, reference outcomes and missingness separately; use review policies instead of treating high scores as facts. Draw your **input**, deciding step, result and limit.

Supported question: Was the unknown label guaranteed correct because its **confidence** was **80%**?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Confidence Is Not Correctness: the case

02. Follow the deciding steps

03. Known outcomes and missing evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.